Diabetes = to pass through ; mellitus = honey (sugar in urine)

Insulin is a hormone produced by the beta cells of the pancreas which helps to transport glucose into the cells and utilize them for energy production in the body.

It also helps to convert glucose to glycogen to be stored in the liver and muscles. Alongside it helps to convert glucose to fat that is stored as body fat and promotes the uptake of amino acids by muscles.

Individuals with diabetes either do not produce insulin or cannot use the insulin produced. As a result glucose does not enter the body cells and builds up in the blood causing an increase in blood glucose levels. As the disease progresses, glucose starts to appear in the urine.

Classification of Diabetes Mellitus

Type 1 Diabetes Mellitus

Type 1 diabetes mellitus accounts for 5% to 10 % of all cases of diabetes. Previously called insulin dependent diabetes is caused by the destruction of the beta cells in the pancreas.

The onset of type 1 diabetes occurs rapidly in children and adolescence. Individuals with type 1 diabetes mellitus require insulin injections for their survival.

Type 2 Diabetes Mellitus

The most common form with approximately 90% to 95% of the individuals belonging to this category. When the body is not producing enough insulin or the insulin it produces cannot be used (insulin resistance) leads to type 2 diabetes mellitus.

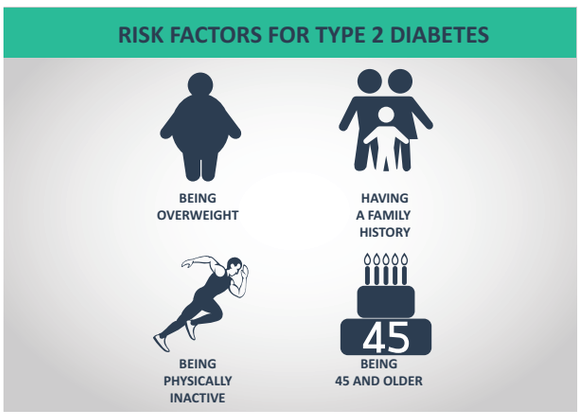
This form has a strong genetic link and is more prevalent in older, obese, inactive individuals. The onset occurs in adults above age 40, but is now also being diagnosed in children and young adults. These individuals do not need insulin for survival, but rather rely on diet, exercise, and (usually) oral medications for disease management.

Gestational Diabetes Mellitus

Gestational diabetes mellitus (GDM) is a temporary form of diabetes developed during pregnancy in woman who do not have diabetes. The blood glucose level become normalized after delivery.

Impaired Glucose Tolerance / Pre-diabetes

Individuals whose fasting blood glucose is above normal (≥ 110 mg/dL), but is not high enough to be diagnosed as diabetes (< 126 mg/dL) fall under this category. It is also referred to as pre-diabetes.



Risk Factors for Type 2 Diabetes Mellitus

- Family history of diabetes.
- Age - 45 years or older.
- Obesity (BMI ≥23 kg/m^2).
- Physical inactivity.

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